

# Differential transcendence of the Hahn–Exton $q$ -cosine and the arithmetic of its zeros

Vico Bonfioli\*

July 17, 2026

## Abstract

Let  $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$ , the Hahn–Exton (third Jackson)  $q$ -cosine. We prove that for every  $0 < q < 1$  the difference Galois group over  $\mathbb{C}(z)$  of the second-order  $q$ -difference equation satisfied by  $G$  contains  $\mathrm{SL}_2$ , and that the equation admits no  $\sigma\delta$ -integrability relation; via the parametrized difference Galois theory of Hardouin–Singer and Dreyfus–Hardouin–Roques it follows that  $G(q, \cdot)$  is differentially transcendental over  $\mathbb{C}(z)$ . We complement this function-level result with an exact evaluation of the regularized product of the positive zeros,  $\prod_{k \geq 1} z_k(q) q^{2(k-1)} = (q; q^2)_\infty$ , a modular value. We then pose, and give effective evidence for, the arithmetic question that these results frame: the transcendence of the zeros  $z_k(q)$  at algebraic  $q$ . This question specializes to the transcendence of a specific reflection-group growth constant  $\beta_2$  (OEIS A396406), for which we prove an unconditional effective non-rationality bound.

## 1 Introduction

The three Jackson  $q$ -analogues of the Bessel function are among the most-studied special functions of  $q$ -series. The third, due to Hahn and Exton, has the  $q$ -cosine and  $q$ -sine specializations

$$G(q, z) := {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} \frac{(-1)^k q^{k(k-1)}}{(q; q)_{2k}} z^k,$$

which is, up to an explicit prefactor,  $J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$ ; it is entire of order 0 in  $z$ , and its positive zeros  $0 < z_1(q) < z_2(q) < \dots$  are real and simple [6]. The analytic theory of these zeros — orthogonality, interlacing, asymptotics — is classical. Their *arithmetic* is, to our knowledge, untouched: nothing is known about the transcendence of  $z_k(q)$  at algebraic  $q$ , the  $q$ -analogue of Siegel’s theorem that the zeros of the Bessel function  $J_\nu$  are transcendental.

This paper establishes the function-theoretic groundwork for that arithmetic and states the resulting transcendence problem precisely. Our results are of three kinds.

*Galois theory (§3).* We compute enough of the difference Galois structure of the defining equation to conclude differential transcendence.

**Theorem 1.** *For every  $0 < q < 1$ , the Hahn–Exton  $q$ -cosine  $G(q, \cdot)$  satisfies no algebraic differential equation over  $\mathbb{C}(z)$ .*

---

\*Independent researcher. [vico@anvilstack.com](mailto:vico@anvilstack.com). Code and verification data: <https://github.com/ElVec1o/pythagorean-reflection-sequence>.

The engine is an unconditional proof that the difference Galois group contains  $\mathrm{SL}_2$  (Theorem 5), obtained by ruling out rank-1 submodules of every symmetric power through a growth estimate on asymptotic-expansion coefficients, together with the absence of a  $\sigma\delta$ -integrability relation (Proposition 6).

A modular zero product (§4). Writing  $Q = q^2$ ,

**Theorem 2.**  $\prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty$ , and  $G(q, z)/\theta_Q(-z) \rightarrow 1/(q; q)_\infty$  along the negative axis, where  $\theta_Q(-z) = \sum_n (-1)^n q^{n(n-1)} z^n$ .

The regularized product of the zeros is thus an explicit modular value — transcendental at algebraic  $q$  by Nesterenko's theorem [7].

Arithmetic of the zeros (§5). These results frame a clean conjecture.

**Conjecture 3 (Z).** For algebraic  $q \in (0, 1)$ , every positive zero  $z_k(q)$  of  $G(q, \cdot)$  is transcendental.

We explain why Conjecture 3 lies just past the reach of the  $q$ -analogue of the Siegel–Shidlovskii method (the double-Pochhammer denominator  $(q; q)_{2k}$  places it below the Siegel margin of [1]), and we record an unconditional partial result for the specific zero of number-theoretic interest.

The motivating instance is the growth constant  $\beta_2 = 1.4916177871\dots$  of the geodesic growth series of a planar right-triangle reflection group (OEIS A396406), which equals  $1/\sqrt{q^*}$  where  $q^*$  is the least positive root of  $S(q) := G(q, 2q(1-q))$ ; equivalently  $z^* := 2q^*(1-q^*) = z_1(q^*)$  is the smallest positive zero of  $G(q^*, \cdot)$ . Conjecture 3 implies  $\beta_2$  transcendental. We prove unconditionally (Proposition 16) that  $q^*$  has no rational representation of denominator  $\leq 10^{191}$ .

All numerical claims are reproducible from the repository above; each is annotated with the script that verifies it and the precision attained.

## 2 The $q$ -difference module

Throughout  $Q = q^2$ ,  $\sigma f(z) = f(Qz)$ , and  $a(z) = q + 1 - qz$ . From the  ${}_1\phi_1$  recurrence,  $G$  satisfies

$$q G(z) - a(z) G(Qz) + G(Q^2 z) = 0, \quad L := q - a\sigma + \sigma^2, \quad (1)$$

a second-order  $q$ -difference equation with companion matrix  $A(z) = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$ ,  $\det A = q$ , regular (Poincaré-type) at  $z = 0$  and irregular at  $z = \infty$  (Newton slopes  $\{0, 1\}$ ). A second solution is  $\tilde{G}(z) = z^{1/2} H(z)$  with  $H(z) = \sum_{k \geq 0} (-1)^k q^{k^2} (1-q) z^k / (q; q)_{2k+1}$ ; the pair  $(G, H)$  realizes the Hahn–Exton  $q$ -cosine/ $q$ -sine  $J_{\mp 1/2}^{(3)}$ . The Casoratian  $\mathrm{Cas}(G, \tilde{G})(z) := G(z)\tilde{G}(Qz) - G(Qz)\tilde{G}(z) = c z^{1/2}$  is nonzero, and the exact  $q$ -Wronskian identity

$$q G(z) H(Qz) - G(Qz) H(z) = q - 1 \quad (2)$$

holds (verified to 40 digits). We write  $M$  for the associated rank-2  $\sigma$ -module over  $\mathbb{C}(z)$ .

At  $z = \infty$  the equation has a formal solution basis reflecting the two Newton slopes. With  $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$ , these are

$$f_1(z) = \theta_Q(-z) \hat{g}(z), \quad \hat{g}(z) = \sum_{k \geq 0} b_k z^{-k}, \quad b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}; \quad f_2(z) = \frac{h(z)}{\theta_Q(-z/q)}, \quad h(z) = \sum_{k \geq 0} c_k z^{-k},$$

where  $\hat{g}$  is  $q$ -Gevrey divergent (its  $q$ -Borel transform is  $1/(q^2/\xi; q)_\infty$ , so  $f_1$  is Borel–Laplace summable off one Stokes direction) and  $h$  is convergent ( $c_k = O(Q^{k^2/2})$ ). The coefficients  $b_k$ , which drive the Galois argument below, satisfy  $b_k \asymp q^{-k^2/4}$ ; both  $\hat{g}$  and  $h$  are determined by (1) and  $b_0 = c_0 = 1$ . Full details of the resummation are in [3]; here only the asymptotic expansions are used.

### 3 The difference Galois group

**Lemma 4** (irreducibility of the  $\sigma$ -module). *For  $0 < q < 1$  the rank-2  $\sigma$ -module of  $G$  is irreducible over  $\mathbb{C}(z)$ : the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

*has no solution  $u \in \mathbb{C}(z)$ .*

*Proof.* Suppose  $u = P/R$  with  $P, R \in \mathbb{C}[z]$  coprime, of degrees  $p, r$ , leading coefficients  $\alpha, \beta$ . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At  $z = 0$ :  $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$ ; if  $R(0) = 0$  then  $P(0) = 0$ , contradicting coprimality, and symmetrically; so  $P(0)R(0) \neq 0$  and  $u(0) \in \{1, q\}$ . *Top degree*: the three terms have degrees  $2p, p+r+1, 2r$ . If  $p = r$ , the top degree  $2p+1$  is attained by the middle term alone, with coefficient  $q\alpha\beta Q^r \neq 0$ ; if  $|p-r| \geq 2$ , the top degree is attained by the first or third term alone, with coefficient  $\alpha^2 Q^p$  or  $q\beta^2 Q^r$ . Both are impossible, so  $|p-r| = 1$ . *Root pairing*: let  $\zeta \neq 0$  be a root of  $R$  of multiplicity  $m$ , so  $P(\zeta) \neq 0$ . At  $z = \zeta/Q$  the three terms of  $F$  have orders  $k, k+m, m_0+m$ , where  $k = \text{ord}_{\zeta/Q} P$  and  $m_0 = \text{ord}_{\zeta/Q} R$ ; since  $F \equiv 0$  the minimum order cannot be attained by a single term, forcing  $k \geq m_0 + m \geq m$ . Symmetrically, a root  $\xi \neq 0$  of  $P$  of multiplicity  $k$  forces  $\text{ord}_{Q\xi} R \geq k$  (coprimality gives  $\text{ord}_{Q\xi} P = 0$ , so the first term has order  $k$  at  $\xi$ , the third  $\text{ord}_{Q\xi} R$ , and the middle at least  $k + \text{ord}_{Q\xi} R$ ; again no single minimum). The two maps  $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$  are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities,  $p \leq r \leq p$ , so  $p = r$  — contradicting  $|p-r| = 1$ .  $\square$

**Theorem 5** (the difference Galois group contains  $\text{SL}_2$ ). *For every  $0 < q < 1$ , the difference Galois group over  $\mathbb{C}(z)$  of the module of  $G$  contains  $\text{SL}_2$ .*

*Proof.* For  $N \geq 1$  suppose  $\text{Sym}^N M$  has a rank-1 submodule, i.e. there is a nonzero solution  $w$  of the  $N$ -th symmetric-power equation with  $r := w(Q\cdot)/w \in \mathbb{C}(z)$ . The products  $f_1^{N-k} f_2^k$  ( $0 \leq k \leq N$ ) form a solution basis (their Casoratian is a power of  $\text{Cas}(f_1, f_2) \neq 0$ ; checked numerically for  $N = 2$ ), so  $w = \sum_k c_k f_1^{N-k} f_2^k$  with constants  $c_k$ ; set  $k_0 := \min\{k : c_k \neq 0\}$ .

*Case  $k_0 < N$ .* Along a ray in a regular class,  $f_2/f_1$  decays super-polynomially, so the rational function  $r$  has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left( \frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left( \frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The  $j$ -th coefficient of the formal series on the right is  $(N-k_0)b_j Q^{-j}(1 + O(q^{cj}))$  with  $b_j = q^{j-j(j-1)/2}/(q;q)_j$ : the dominant term comes from  $b_j Q^{-j}$  in  $\hat{g}(Qz)/\hat{g}(z)$ , all competing products being smaller by a factor  $q^{cj}$  (verified: the ratio of the  $j$ -th coefficient to  $(N-k_0)b_j(Q^{-j}-1)$  is  $1 - 10^{-6}$  at  $j = 12, q = \frac{1}{3}$ ). These coefficients grow like  $q^{-j^2/2}$ , whereas a rational function's expansion at  $\infty$  has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational  $r$  exists.

*Case  $k_0 = N$ .* Then  $r = (-z/q)^N \rho^N$  with  $\rho := h(Q\cdot)/h$  meromorphic on  $\mathbb{C}^*$ ;  $\rho^N \in \mathbb{C}(z)$  and single-valuedness force  $\rho \in \mathbb{C}(z)$  (an  $N$ -th root of a rational function is single-valued on  $\mathbb{C}^*$  only if every divisor multiplicity, including at 0 and  $\infty$ , is divisible by  $N$ ). Then  $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$  is rational, giving a rational solution of the rank-2 Riccati equation — contradicting Lemma 4.

Hence no  $\text{Sym}^N M$  has a rank-1 submodule.  $N = 1$  is irreducibility;  $N = 2$  excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup  $(A_4, S_4, A_5)$  fixes a line in

some  $\text{Sym}^N$  with  $N \leq 12$  (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of  $\text{GL}_2$ , the group contains  $\text{SL}_2$ .  $\square$

**Proposition 6** (no  $\sigma\delta$ -integrability). *For every  $0 < q < 1$  there is no  $B \in \text{gl}_2(\mathbb{C}(z))$  and no  $c \in \mathbb{C}(z)$  with  $\theta(A) = \sigma(B)A - AB + cA$ , where  $A = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$  and  $\theta = z\partial_z$ .*

*Proof.* Writing  $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ , the entries give  $\gamma = -q\beta_1$ ,  $\delta = \alpha_1 + a\beta_1 - c$ , and eliminating  $\alpha$  (the central terms  $c$  cancel identically; verified symbolically) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$ ,  $a = a(z)$ ,  $a_1 = a(Qz)$ . A polynomial part of degree  $m \geq 0$  contributes top coefficient  $\propto \beta_m(1 - q^{2+2m}) \neq 0$  at degree  $m + 3$ , above the inhomogeneity: so  $\beta$  has no polynomial part. If  $\beta$  has a pole off the orbit  $z_a Q^{\mathbb{Z}}$ , the maximal-modulus pole appears in exactly one term (the slot  $z = \zeta Q^{-3}$ , coefficient  $qa \neq 0$  there), and the minimal-modulus pole likewise (slot  $z = \zeta'$ , coefficient  $qa_1$ ): both impossible; so the poles lie in the window  $\{z_a Q^{-1}, \dots, z_a Q^3\}$  ( $a(z_a Q^k) = (q+1)(1 - Q^k)$  vanishes only at  $k = 0$ ). Sweeping the slots  $z = z_a Q^{-4}, \dots, z_a Q^{-1}$  kills the orders at  $z_a Q^{-1}, \dots, z_a Q^2$  one by one (each pole appears in a single term with nonvanishing coefficient), and the order at  $z_a Q^3$  is killed at the slot  $z = z_a Q$  — coefficient  $a_1(aa_1 - q)$  with  $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$  — or, where that vanishes (one root  $q \approx 0.78027$ ), at the slot  $z = z_a Q^2$  — coefficient  $a(aa_1 - q)$  with  $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$  (one root  $q \approx 0.86753$ ); the two polynomials are coprime, so for every  $q \in (0, 1)$  at least one slot applies. Hence  $\beta \equiv 0$ , contradicting  $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \neq 0$ . (Cross-check: at  $q = \frac{1}{3}$  the exact linear system for poles of order  $\leq 3$  in the window has  $\text{rank}(A) = 15$ ,  $\text{rank}[A|b] = 16$ .)  $\square$

**Corollary 7** (differential transcendence of  $G$ ). *For every  $0 < q < 1$  the function  $G(q, \cdot)$  satisfies no algebraic differential equation over  $\mathbb{C}(z)$ : the Hahn–Exton  $q$ -cosine is differentially transcendental.*

*Proof, via parametrized difference-Galois theory.* We apply the parametrized  $(\sigma, \theta)$ -difference Galois theory with derivation  $\theta = z\partial_z$ . The determinant equation  $\sigma(y) = \det(A)y = qy$  has the solution  $y = z^{1/2}$  (indeed  $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$ , as  $Q = q^2$ ), which is algebraic over  $\mathbb{C}(z)$ ; hence the parametrized difference Galois group of  $\sigma(y) = \det(A)y$  lies in  $\mathbb{C}^\times$ . This is exactly the hypothesis of Dreyfus–Hardouin–Roques [4, Prop. 2.9], which for a rank-2 system whose difference Galois group contains  $\text{SL}_2$  (here Theorem 5) gives the dichotomy: the parametrized group  $\text{Gal}^\theta$  either is conjugate into  $\text{GL}_2(\mathbb{C})$  (all solutions differentially algebraic) or contains  $\text{SL}_2(\tilde{\mathbb{C}})$  (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists  $B \in \text{gl}_2(\mathbb{C}(z))$  with  $\theta(A) = \sigma(B)A - AB$  (their integrability condition (7);  $\theta(\det A) = 0$  since  $\det A = q$  is constant in  $z$ , so no trace term appears). By Proposition 6 no such  $B$  exists — indeed none exists even allowing a scalar twist  $cA$ . Therefore the second alternative holds:  $\text{Gal}^\theta \supseteq \text{SL}_2(\tilde{\mathbb{C}})$ , and every nonzero solution, in particular  $G(q, \cdot)$ , is differentially transcendental over  $\mathbb{C}(z)$ .  $\square$

## 4 The connection constant and the zero product

The behaviour of  $G$  at  $z = \infty$  is governed by a theta-comparison. Let  $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$ , so  $\theta_Q(Qx) = x^{-1}\theta_Q(x)$  and, by the Jacobi triple product,  $\theta_Q(-z) = (Q; Q)_\infty(z; Q)_\infty(Q/z; Q)_\infty$ , an entire function of order 0 with simple zeros exactly at  $z \in Q^{\mathbb{Z}}$ . The following theorem evaluates the leading asymptotic constant of  $G$  and, as a consequence, the regularized product of its zeros; the constant  $C_1 := \lim_{R \rightarrow \infty} G(-R)/\theta_Q(R)$  coincides with the Stokes-invariant connection coefficient of  $G$  against the theta solution.

**Theorem 8** (the connection constant, and the zero product of the Hahn–Exton  $q$ -cosine). For  $0 < q < 1$ ,

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where  $z_1 < z_2 < \dots$  are the positive zeros of  $G(q, \cdot)$ . Likewise, for the second solution's underlying entire function  $H$  with zeros  $w_1 < w_2 < \dots$ ,  $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$ . (Verified at three bases including  $q^*$ , to 41, 23, 26 digits for  $C_1$  and 58 digits for the product.)

*Proof.* (1) *An exact identity.* From  $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$  and Euler's expansion  $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$  at  $x = q^{2k+1}$ , together with  $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$  and the substitution  $m = k + j$ , absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)/2} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the  $c_j$  coincide with the  $b_j$  of §2; identity checked symbolically through  $z^8$  and numerically to 41 digits).

(2) *The constant.* Fix  $u < 0$  and put  $z = uQ^{-N}$ . The moduli  $q^{m(m-1)/2}|z|^m$  peak at  $m_0 = N + O(1)$  and decay like  $q^{(m-m_0)^2 + O(1)}$  away from it. For  $1 \leq j \leq m \leq 2N$ ,  $2Nj - j(j-1)/2 \geq Nj$ , so  $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$  and  $|P_m(1/z) - 1| \leq C'q^N$  uniformly; the terms with  $m > 2N$  contribute a relative  $O(q^{N^2})$ . The negative- $m$  part of  $\theta_Q(-z)$  is  $O(|z|^{-1})$ , negligible against the peak. Hence  $(q; q)_\infty G(z) / \theta_Q(-z) \rightarrow 1$  along  $z = uQ^{-N}$ , for each fixed  $u < 0$ .

(3) *Geometric zero dressing.* On the circle  $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$  the identity of step (1) and the bounds of step (2) give  $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$ , while  $\theta_Q(-z)$  has a simple zero at  $Q^{1-k}$  with derivative bounded below there; Rouché gives  $z_k = Q^{1-k}(1 + O(q^k))$ , i.e.  $\log(z_k Q^{k-1}) = O(q^k)$ , absolutely summable.

(4) *The product.*  $G(z) = \prod_k (1 - z/z_k)$  (order 0, genus 0,  $G(0) = 1$ ) and  $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$  (Jacobi). Along  $z = -R$ ,  $R \rightarrow \infty$ ,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for  $x, y > 0$  the function  $R \mapsto \log \frac{1+Rx}{1+Ry}$  is monotone from 0 to  $\log(x/y)$ , so each factor is dominated by  $|\log(z_k Q^{k-1})|$ , summable by step (3). Dominated convergence gives  $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$  as  $R \rightarrow \infty$ ; evaluating along the subsequence  $R = |u|Q^{-N}$  by step (2), the limit equals  $1/(q; q)_\infty$ , whence  $C_1 = 1/(q; q)_\infty$  and  $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$  by Euler's  $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$ . The  $H$ -statement is identical with comparison function  $\theta_Q(-qz)$  and ratio  $(1 - q)/(q; q)_\infty$ .  $\square$

*Remark 9* (the modular split at  $z^*$ ). At  $q = q^*$  the product splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.988299884628746\dots \quad (\text{verified to 30 digits}).$$

Were  $z^*$  (equivalently  $\beta_2$ ) algebraic, the modular value  $(q; q^2)_\infty$  — transcendental at algebraic  $q$  by Nesterenko's theorem [7] applied to the Euler products — would equal an algebraic number times the tail product  $\prod_{k \geq 2} z_k Q^{k-1}$ . This is consistent only if the tail product is itself transcendental; the arithmetic of the zero condition is thereby confined to that tail, which is the content of the multiplicity problem of [3].

## 5 The arithmetic of the zeros

### 5.1 Conjecture Z and its confluent evidence

Conjecture 3 is the  $q$ -analogue of the classical theorem (Siegel; see also Lindemann for the endpoint) that the positive zeros  $j_{\nu,k}$  of the Bessel function  $J_\nu$  are transcendental for rational  $\nu$ . Two structural facts support it. First, Theorem 1:  $G$  is differentially transcendental, so its zeros are not the zeros of any algebraic-differential shadow whose arithmetic could be controlled classically. Second, in the confluent limit  $q \rightarrow 1$  the equation (1) degenerates to Bessel's equation and  $z_k(q) \rightarrow (qj_{-1/2,k})^2$ , so Conjecture 3 deforms Siegel's theorem.

### 5.2 The self-referential specialization: $\beta_2$

The reflection-group growth constant  $\beta_2$  is the reciprocal square root of the least positive root  $q^*$  of  $S(q) = G(q, 2q(1 - q))$ , which is exactly the condition that the catalytic point  $z^* = 2q^*(1 - q^*)$  be a zero of  $G(q^*, \cdot)$ . Thus  $\beta_2$  is governed by a *self-referential* specialization of Conjecture 3: the zero and the base move together along the diagonal  $z = 2q(1 - q)$ .

At this point the  $q$ -Wronskian (2) yields a clean exact relation between two Hahn–Exton values: since  $G(q^*, z^*) = 0$ ,

$$G(q^*, q^{*2}z^*) \cdot H(q^*, z^*) = 1 - q^* \quad (3)$$

(verified to 25 digits) — a product of two  $q$ -Bessel values at the algebraic point  $z^*$  equal to the algebraic number  $1 - q^*$ . Together with  $G(q^*, z^*) = 0$  and the modular zero product of Theorem 2, this places three exact constraints on the values at  $z^*$ ; a transcendence proof requires one more independent constraint than the reductio supplies, which is the precise sense in which Conjecture 3 is a multiplicity (rather than a single-relation) problem.

### 5.3 The $q$ -Siegel–Shidlovskii margin

The natural route to Conjecture 3 is the  $q$ -analogue of the Siegel–Shidlovskii method developed by Amou–Matala-aho–Väänänen [1]. That method applies to Poincaré-type  $q$ -difference systems — a hypothesis (1) satisfies at  $z = 0$  — under a *Siegel margin*  $\gamma < \Gamma$ , where  $\gamma$  is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class this ratio is

$$\gamma = \frac{\log(t/s)}{2 \log t} \leq \frac{1}{2} \quad (q = s/t \text{ in lowest terms}),$$

the factor 2 being the double Pochhammer  $(q; q)_{2k}$ ; this is below the threshold, so the method does not apply as stated. Extending it past the margin for this irregular-at- $\infty$  class is the outstanding problem; the connection data needed for the irregular direction is available in closed form [3].

### 5.4 Integrality of the zero curve

The smallest zero, viewed as a function of the base, is an integer power series.

**Theorem 10** (integrality of the zero curve). *The branch  $z_1(q)$  of the smallest positive zero of  $G(q, \cdot)$  with  $z_1(0) = 1$  satisfies*

$$z_1(q) = 1 - q - q^3 + q^4 - q^5 + 2q^6 - 2q^7 + 4q^8 - 6q^9 + 8q^{10} - 14q^{11} + 21q^{12} - \dots \in \mathbb{Z}[[q]].$$



*Proof.* Each coefficient  $g_k = (-1)^k q^{k(k-1)} / (q; q)_{2k}$  lies in  $\mathbb{Z}[[q]]$  (the reciprocal Pochhammer is the generating function of partitions into parts  $\leq 2k$ ), and  $g_k \equiv 0 \pmod{q^{k(k-1)}}$ , so  $G(q, z) = \sum_k g_k z^k$  converges  $(q)$ -adically for  $z \in 1 + q\mathbb{Z}[[q]]$  and reduces at  $q = 0$  to  $G(0, z) = 1 - z$ . Since  $G(0, 1) = 0$  and  $G_z(0, 1) = -1$  is a unit, Hensel's lemma in the complete local ring  $\mathbb{Z}[[q]]$  yields a unique root  $z_1 \in 1 + q\mathbb{Z}[[q]]$ . (Verified to order  $q^{56}$ ; the coefficient signs alternate from  $n = 3$  on.)  $\square$

*Remark 11* (arithmetic and analytic features of the zero curve). (a) The series is not rational and satisfies no algebraic relation  $P(q, z_1) = 0$  of bidegree  $\leq (6, 5)$  (exact linear algebra on 57 coefficients). (b) Its radius of convergence is  $\approx 0.5696$  ( $|c_{56}/c_{55}| = 1.7557$ ), limited by a complex branch point of the zero configuration; in particular  $q^* = 0.4494\dots$  lies inside the disc of convergence. (c) The integrality opens a *linear-height* approximation ledger: at  $q = s/t$  the  $N$ -th partial sum is rational with denominator  $t^N$  and error  $\asymp (q/R)^N$ ,  $R \approx 0.5696$ , so the Liouville criterion for the crossing  $z_1(q) = 2q(1 - q)$  reads  $s < R$  — failing, but now only by the factor  $s/R \approx 1.76$  at  $s = 1$ , the narrowest margin among all ledgers recorded for this problem; the obstruction has become the sub-unit radius, i.e. the complex collision of zeros. (d) The integer coefficient sequence  $1, -1, 0, -1, 1, -1, 2, -2, 4, -6, 8, -14, 21, -34, 56, -88, 148, -242, 398, -669, 1109, \dots$  appears to be new.

## 5.5 The second-kind forms and the exact reduction

The  $q$ -Siegel construction for this class is explicit. For  $N \geq 1$  put

$$P_N(q, z) := (q; q)_{2N} \sum_{k=0}^N \frac{(-1)^k q^{k(k-1)} z^k}{(q; q)_{2k}} = \sum_{k=0}^N (-1)^k q^{k(k-1)} z^k \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q, z].$$

**Proposition 12** (the second-kind forms).  $\deg_q P_N = 2N^2 + N$  and  $\deg_z P_N = N$  (exact; verified  $N \leq 4$ ), and the remainder  $R_N(q, z) := (q; q)_{2N} G(q, z) - P_N(q, z)$  satisfies, uniformly for  $q$  in compacts of  $(0, 1)$  and  $z$  bounded,

$$|R_N(q, z)| = q^{N^2(1+o(1))}.$$

*Proof.* Membership in  $\mathbb{Z}[q, z]$  and  $\deg_z P_N = N$  are clear from the displayed product form, since  $(q; q)_{2N} / (q; q)_{2k} = \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q]$ . For the  $q$ -degree, the  $k$ -th summand has degree  $k(k-1) + \sum_{i=2k+1}^{2N} i = 2N^2 + N - k^2 - 2k$ , maximal at  $k = 0$  with value  $\deg(q; q)_{2N} = 2N^2 + N$ ; the maximum is attained once, so no cancellation occurs. For the remainder,  $R_N = (q; q)_{2N} \sum_{k>N} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$ , whose leading term ( $k = N+1$ ) is  $\pm q^{N(N+1)} z^{N+1} (q; q)_{2N} / (q; q)_{2N+2}$ ; the tail is dominated by a geometric series in  $q^{2N}$ , so  $|R_N| = q^{N^2+O(N)}$  uniformly. (Measured  $\log |R_N| / N^2 \rightarrow \log q$  at both  $z = 2q(1 - q)$  and a generic  $z_0$ .)  $\square$

**Proposition 13** (reduction to the Siegel margin). Let  $q = s/t \in (0, 1)$  and let  $z_0 = a/b \in \mathbb{Q}^\times$  with  $G(q, z_0) = 0$ . Then  $P_N(q, z_0) = -R_N(q, z_0)$  and  $t^{2N^2+N} b^N P_N(q, z_0) \in \mathbb{Z}$ . Consequently, if for infinitely many  $N$  one has  $P_N(q, z_0) \neq 0$  and

$$|R_N(q, z_0)| < t^{-(2N^2+N)} b^{-N},$$

then  $G(q, z_0) \neq 0$  — a contradiction. By Proposition 12 the displayed criterion reads  $q^{N^2} < t^{-2N^2}$  up to  $O(N)$  in the exponent, i.e.

$$\log(t/s) > 2 \log t \iff st < 1,$$

which is false. The deficit is exactly the factor 2 in  $\deg_q P_N = \deg(q; q)_{2N}$ : the double Pochhammer.

*Proof.*  $P_N(q, z_0) = (q; q)_{2N}G(q, z_0) - R_N(q, z_0) = -R_N(q, z_0)$  by hypothesis; clearing denominators of  $P_N \in \mathbb{Z}[q, z]$  at  $q = s/t$ ,  $z = a/b$  gives  $t^{\deg_q P_N} b^{\deg_z P_N} P_N(q, z_0) \in \mathbb{Z}$ , and a nonzero integer has modulus  $\geq 1$ . The stated evaluation of the criterion is Proposition 12.  $\square$

**Corollary 14** (what would suffice). *Conjecture 3 at  $z_0$  — hence, on the diagonal, the irrationality of  $\beta_2$  — follows from any family of forms  $\Lambda_N \in \mathbb{Z}[q, z]$  with  $\Lambda_N(q, z_0) \neq 0$ , remainder  $q^{N^2(1+o(1))}$  against  $G$ , and*

$$\deg_q \Lambda_N \leq N^2 \frac{\log(t/s)}{\log t} (1 - \varepsilon), \quad \text{in particular } \deg_q \Lambda_N \leq N^2(1 - \varepsilon) \text{ when } s = 1.$$

The forms of Proposition 12 have  $\deg_q = 2N^2$ : it suffices to halve the  $q$ -degree at the same remainder, or equivalently to double the remainder decay to  $q^{2N^2}$ . A single-Pochhammer clearing  $(q; q)_N$  would have degree  $\frac{1}{2}N^2(1+o(1))$  and would suffice at once; no such representation of  $G$  along a non- $q$ -power argument exists in the Rogers–Ramanujan/Bailey framework.

*Remark 15* (the theta-decay barrier: why Corollary 14 is unreachable by truncation). The hypothesis of Corollary 14 cannot be met by *any* truncate-and-clear form. Let  $\sum_k c_k(q) q^{\alpha k^2} z^k$  ( $\alpha > 0$ ,  $c_k \in \mathbb{Q}(q)$ ) be truncated at  $N$  and cleared. The  $k = N$  summand already contributes  $q^{\alpha N^2}$ , so  $\deg_q \geq \alpha N^2$ ; the remainder is the first omitted term, of size  $q^{\alpha N^2(1+o(1))}$ . The criterion  $|R_N| < t^{-\deg_q}$  therefore reads

$$\alpha N^2 \log(t/s) > \alpha N^2 \log t \iff s < 1,$$

false for every  $q = s/t \in (0, 1)$ : the decay and the degree are driven by the same factor  $q^{\alpha k^2}$  and cancel exactly. Denominators only make matters worse. Numerically, the cleared truncations have  $\deg_q = N^2$  exactly for the theta series  $\sum q^{k^2} z^k$  and for the Rogers–Ramanujan series  $\sum q^{k^2} / (q; q)_k$  (both  $\beta := \deg_q / N^2 = 1$ , the boundary), while the Hahn–Exton series sits at  $\beta \rightarrow 2$  (2.25, 2.17, 2.12 at  $N = 4, 6, 8$ ) — a further factor 2 past a boundary that is itself already unreachable.

This explains the shape of the known landscape rather than merely our position in it: the transcendence of theta values and of Rogers–Ramanujan values — both at  $\beta = 1$  — is known *only* through Nesterenko’s modular method [7], never through a Siegel/Liouville margin, precisely because no theta-decay series can cross  $\beta < 1$ . Consequently the only mechanisms available for Conjecture 3 are modularity, which  $G$  provably lacks (it is off the modular class; see [3]), or a multiplicity theory for the associated  $(\sigma, \delta)$ -module. The double Pochhammer is not the whole obstruction — it is a factor 2 on top of a structural one.

## 5.6 An unconditional effective bound

Although Conjecture 3 is open, the specific root  $q^*$  admits an effective non-rationality statement, proved by certified interval arithmetic.

**Proposition 16** (effective non-rationality of  $q^*$ ). *The least positive root  $q^*$  of  $S$  admits no rational representation  $s/t$  with  $t \leq 10^{191}$ . Consequently, if  $\beta_2 = u/v$  in lowest terms then  $u > 10^{95}$ .*

*Proof.* All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational`) with the series tail bounded by  $(q; q)_{2k} \geq (1 - 2q)/(1 - q) > 0$  for  $q < \frac{1}{2}$  and term ratios  $< \frac{1}{2}$  beyond the truncation. (i)  $S > 0$  on  $[0, 0.40]$  (400 certified chunks). (ii)  $S' \leq -1$  on  $[0.40, 0.46]$  (600 certified chunks;  $S'$  evaluated term-by-term with the logarithmic-derivative weights), so  $S$  has at most one zero in  $[0.40, 0.46]$ . (iii) with  $a = 0.d_1 \cdots d_{385}$  the 385-digit truncation of  $q^*$  and  $b = a + 2 \cdot 10^{-385}$ , certified evaluation gives  $S(a) > 0 > S(b)$ ; by (i)–(ii),  $q^* \in [a, b]$ . (iv) the interval



$[a, b]$  contains the rational  $s_0/t_0$  (Stern–Brocot minimal fraction) with  $t_0 \geq 10^{192}$ , verified by exact rational comparison, and certified  $S(s_0/t_0) \neq 0$ . If  $m/n \in [a, b]$  with  $n \leq 10^{191}$  and  $m/n \neq s_0/t_0$ , then  $|m/n - s_0/t_0| \geq 1/(nt_0) \geq 10^{-384} > b - a$ , impossible; so  $m/n = s_0/t_0$ , whence  $n = t_0 > 10^{191}$ , a contradiction. Thus no rational with denominator  $\leq 10^{191}$  lies in  $[a, b]$  at all, in particular none equals  $q^*$ . For  $\beta_2 = u/v$  in lowest terms,  $q^* = v^2/u^2$  in lowest terms, so  $u^2 > 10^{191}$ .  $\square$

## References

- [1] M. Amou, T. Matala-aho, K. Väänänen, *On Siegel–Shidlovskii’s theory for  $q$ -difference equations*, Acta Arith. **127** (2007), 309–335.
- [2] C. E. Arreche, T. Dreyfus, J. Roques, *Differential transcendence criteria for second-order linear difference equations and elliptic hypergeometric functions*, J. Éc. polytech. Math. **8** (2021), 147–168.
- [3] V. Bonfiori, *Transcendence of a planar reflection–group growth series and its relaxed companion*, companion manuscript (2026); code and data at the repository above.
- [4] T. Dreyfus, C. Hardouin, J. Roques, *Hypertranscendence of solutions of Mahler equations*, J. Eur. Math. Soc. **20** (2018), 2209–2238.
- [5] C. Hardouin, M. F. Singer, *Differential Galois theory of linear difference equations*, Math. Ann. **342** (2008), 333–377.
- [6] H. T. Koelink, R. F. Swarttouw, *On the zeros of the Hahn–Exton  $q$ -Bessel function and associated  $q$ -Lommel polynomials*, J. Math. Anal. Appl. **186** (1994), 690–710.
- [7] Yu. V. Nesterenko, *Modular functions and transcendence questions*, Sb. Math. **187** (1996), 1319–1348.